Chemical Engineering Journal 465 (2023) 142793

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

An environmental and economic sustainability assessment of coffee production in the UK

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ARTICLE INFO

Keywords:
Process modelling
Life cycle assessment
Life cycle costing
Ground coffee
Instant coffee
Food and drink

ABSTRACT

With around two billion cups consumed daily worldwide, the coffee industry faces pressures to improve its sustainability. Focusing on the UK as the fifth largest coffee consumer in Europe, this paper presents for the first time a life cycle environmental and economic sustainability assessment of coffee production at the product and sectoral levels. Both gate-to-gate and cradle-to-gate system boundaries are considered to identify opportunities for improvements in the manufacturing process and along the supply chain. Process design and simulation are combined with life cycle assessment and life cycle costing to evaluate the sustainability of the production of ground and instant coffee roasted to different levels. Based on the functional unit of 1 tonne of coffee produced, instant coffee has higher impacts and costs than ground coffee for both system boundaries because it requires double the amount of green coffee beans and 7–11 times more energy. For gate-to-gate, roasting and packaging are the main contributors to the impacts and costs; for cradle-to-gate, cultivation of coffee beans and packaging are the hotspots. Light roasts generally have lower impacts and costs due to lower material losses and energy requirements. At the sectoral level, the coffee industry emits 863–894 kt CO_2 eq. annually, contributing 1.3% to the GHG emissions from the UK food and drinks sector. Key opportunities for improvements include switching to lighter roast and pouch packaging, as well as using air recirculation roasters and silverskin as fuel. These results will be of interest to the coffee industry, policy makers and consumers.

1. Introduction

Coffee is considered the most popular drink in the world, with around 10 Mt of coffee consumed every year, or around two billion cups a day [1,2]. Around 33% of global coffee consumption occurs in Europe; Asia and Oceania are the next largest consumers with a market share of 22%, followed by North America (19%), South America (16%), Africa (7%), and Central America and Mexico (3%) [2]. Approximately 95 million cups are consumed daily in the UK, making it the fifth largest market in Europe [1,3]. The annual turnover of the UK coffee industry is estimated at £7.2 billion, with an average growth of 7% in the past six years [4]. Around 181,000 tonnes of green coffee beans are imported to the UK every year to be processed and consumed. As indicated in Table 1, 66% of this comes from Brazil, Vietnam, Colombia and Honduras, the main coffee growers in the world [3]. The UK also imports roasted coffee from EU countries, such as Germany, Italy and Spain, amounting to around 20% of the total national coffee imports.

Different varieties of coffee products are available in the UK market and can be divided into five main categories [4]. According to their volume sales reported in 2017 these are [4]:

(i) regular instant standard coffee (48%);
(ii) standard fresh ground coffee (24%);
(iii) fresh ground coffee pods (16%);
(iv) instant-coffee mixes (4%); and
(v) decaffeinated instant standard coffee (8%).

The first four catagories account for the large majority (92%) of the total national sales by volume, equivalent to 73 kt, with decaffeinated coffee totalling 6 kt of sales [4]. In the UK, coffee sales and turnover have increased by 20% and 27%, respectively, between 2012 and 2017 [4], a trend that is likely to continue due to the increasing demand. For instance, the demand for coffee pods has quadrupled their sales volume and turnover in six years [4].

However, coffee is considered one of the most energy-intensive food products. For example, a European Commission study [5] estimated the life cycle embedded energy of coffee at around 230 MJ/kg, with the processing stage contributing 33% to the total energy consumption. This

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https://doi.org/10.1016/j.cej.2023.142793
Received 12 January 2023; Received in revised form 8 March 2023; Accepted 2 April 2023
Available online 5 April 2023
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Table 1
Annual average coffee imports to the UK [13,14].

Imported coffee product	Country of origin	Quantity[a] (kt)
Green coffee beans	Brazil	51.5
	Vietnam	36.9
	Colombia	19.1
	Honduras	12.9
	Others	60.8
Roasted coffee	Germany	12.1
	Italy	10.4
	Spain	6.5
	France	4.6
	Netherlands	4.5
	Others	11.2

[a] Based on average imported values from 2014 to 2019 [13].

is mostly due to the roasting process which takes place at temperatures between 190 °C and 245 °C, depending on the roasting level [6]. In the UK, the coffee industry is responsible for 3.2% of the total energy consumed by the food and drink sector [7]. Furthermore, the production of coffee contributes to emissions that represent potential threats to human health and the environment, such as carbon monoxide, carbon dioxide, particulate matter and volatile organic compounds [8,9]. In an effort to reduce the energy consumption of the food and drink sector, the UK government [10-12] has launched the industrial decarbonisation and energy efficieny road map action plan which aims to increase the sector's energy efficiency while reducing the greenhouse gas emissions. This will enable the different industries within the sector to develop strategies for improving energy efficiency acrosss their different unit operations, such as roasting and extraction in the case of the coffee industry [12].

Various efforts have been made to quantify the environmental impacts of coffee, with several life cycle assessment (LCA) studies of coffee reported in the literature, dating from 2003 and spanning seven geographical locations. These are the EU [15,16], Finland [17], Germany [18], Japan [19], US [20], North America [21] and Thailand [22]. Whilst these studies have aimed to understand the environmental impacts of different types of coffee, such as spray-dried instant coffee [15,16,19], coffee pods or capsules [15,19,20], drip-filter coffee [15,19,23], espresso-based drinks, e.g. latte, cappuccino [16,19,23] and canned coffee [19], they have focused on the consumption of coffee. The system boundaries considered were from cradle to grave and cradle to consumer, with most functional units defined as "one serving of coffee" [16,19,21]. Other functional units included "2000 cups of coffee" [18], and "1 L of consumed coffee" [17]. All studies assessed climate change, and the reported results ranged from 48.6 to 330 g CO_2 eq./serving, with milk-based coffee, such as latte, incurring the highest impact. The cultivation of green coffee beans has been identified as a hotspot across all these studies, contributing 11–85% to the impacts, and the processing stage is reported to cause between <1% and 10% of the total impacts. However, not all studies reported direct contributions from this life cycle stage, thus making it difficult to understand fully the environmental implications from coffee production. All studies relied on average data collected from coffee manufacturing factories and estimations from literature to assess the environmental impacts associated with coffee production, with only three studies being specific about the roasting level. For example, Hick [20] and Phrommarat [22] looked at medium- and Hassard et al. [19] considered dark-roast coffee. Unlike other drinks, coffee production involves various operating conditions and processing stages, depending on the roasting degrees, (light to dark roast), type of product (ground, instant) and manufacturing methods (spray- or freeze-dried instant coffee) [24-26]. These variations will result in different energy consumption requirements and, therefore, should be considered in the inventory data used to assess and improve the environmental sustainability of coffee production. For example, having access to more detailed information can help set more realistic energy-reduction targets [10,11,27].

As for the economic sustainability of coffee production, besides a news article published in the Financial Times [28] that provided a cost breakdown for a cup of coffee, no other economic analysis of coffee production has been reported in the literature. According to that article, a typical cup of coffee is reported to cost £2.50 in the UK, with the large majority of these costs related to the consumption stage of coffee, specifically the overheads and operating costs of a coffee shop. Coffee production, on the other hand, contributes 4% to the total costs (10 pence), with coffee roasting being the major (80%) source of coffee production costs. However, no further details on these costs were reported.

In the light of these knowledge gaps related to coffee production, this study assesses for the first time the environmental and economic sustainability of coffee production to help identify opportunities for improvements. For these purposes, process design and simulation tools have been coupled with LCA and life cycle costing (LCC). The focus is on two main types of coffee available on the market – ground and instant – considering eight different products and four roast levels (from light to dark). Although the study is situated in the UK, the approaches used and the trends of the results obtained here will be applicable to other countries and regions. The findings from this work will be of relevance to coffee producers, consumers and policy makers, as well as researchers in the food sector.

2. Methods

Process design and simulation has been carried out using Aspen Plus V8.8 [29] to develop flowsheets and obtain mass and energy data for the LCA and LCC. The LCA follows the ISO 14040/44 methodology [30,31] and the GaBi V8.6 LCA software [32] has been used to model the systems and estimate the environmental impacts. The LCC is based on the methodologies in Swarr et al. [33] and ISO 15686 [34], and is applied in congruence with the LCA methodology. The next section describes the goal and scope of the study, followed by inventory data and an overview of the impact assessment and LCC methods.

2.1. Goal and scope

The main goal of this study is to quantify the life cycle environmental impacts and costs of ground and instant coffee production and identify opportunities for improvements. Two system boundaries are considered (Fig. 1). The first is from gate-to-gate, focusing on the coffee manufacturing process to help understand better its impacts and costs and identify opportunities for improvements. The main reason for considering this system boundary is that producers can influence the operating conditions of the production process, the choice of energy and other utility sources, as well as packaging materials. Hence, by helping them identify the hotspots, the study aims to inform producers on how to produce coffee more sustainably. In this system boundary, the first "gate" corresponds to the point where green coffee beans leave the plantation and the second "gate" is the coffee factory gate, where the packaged coffee product is ready to leave the manufacturing plant. The activities included are energy and other utilities used in coffee production, packaging materials and waste management (Fig. 1).

The second system boundary is from cradle-to-gate which, in addition to the gate-to-gate activities, includes the cultivation and transportation of green coffee beans from their country of origin to the factory, in this case assumed to be in the UK (Fig. 1). This system boundary is aimed at understanding the level of sustainability and the hotspots in the coffee supply chain that should be targeted for improvements.

At the product level, only instant and fresh ground coffee are considered, excluding coffee pods and instant-coffee mixes, in order to compare the two types of coffee production processes. Coffee pods, which contain ground coffee, are considered in a packaging sensitivity

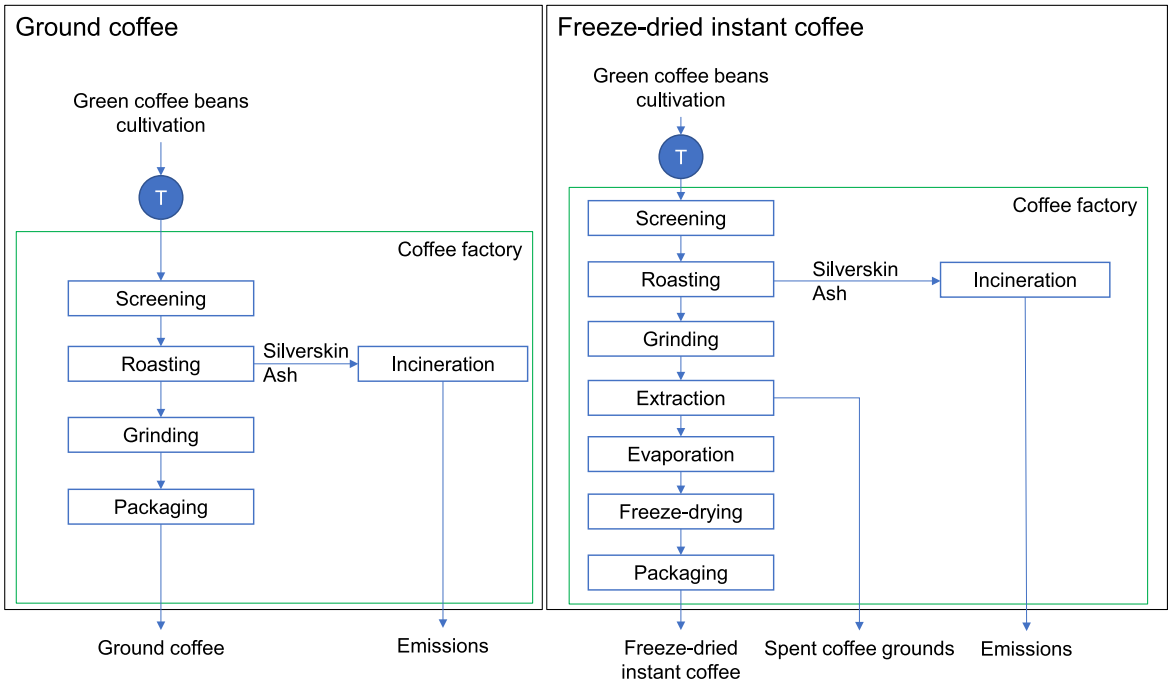


Fig. 1. System boundaries considered in this study for the production of ground and instant coffee. [The inner (green border) box titled "Coffee factory" delineates the gate-to-gate system boundary, while the outer (black border) box represents the cradle-to-gate boundary. T: Transport]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

analysis and at the sectoral level, along with instant-coffee mixes. Decaffeinated coffee is excluded at both the product and sectoral levels due to a relatively small contribution to the total coffee production volumes (8%). The construction of the coffee production plant is also excluded due to a lack of data. Due to long lifetimes (>20 years), the contribution of manufacturing plants to the impacts and costs per functional unit is typically low [35]; hence, this exclusion is considered reasonable.

Four roasting levels have been considered for both ground and instant coffee to reflect the coffee products available on the market. These are light, medium, medium-dark and dark roast, where the green coffee beans are roasted at 192.5 °C, 215 °C, 227.5 °C and 245 °C, respectively [6].

Therefore, the following eight coffee products are considered:

i) light roast ground coffee;
ii) light roast instant coffee;
iii) medium roast ground coffee;
iv) medium roast instant coffee;
v) medium-dark roast ground coffee;
vi) medium-dark roast instant coffee;
vii) dark roast ground coffee; and
viii) dark roast instant coffee.

The functional unit for both system boundaries is defined as the "production of 1 tonne of ground or instant coffee". The cradle-to-gate results are also scaled up to estimate the impacts in the UK coffee sector, including the production of ground coffee pods and instant-coffee mixes. For this purpose, the functional unit is defined as the "annual production of ground and instant coffee products in the UK".

As can be observed in Fig. 1, no co-products are obtained in this process and the valorisation of waste, such as silverskin or chaff and spent coffee grounds (SCGs), has not been considered as it falls outside the scope of the study. Energy recovery from silverskin is explored in a sensitivity analysis in Section 3.2.3.2, as this waste is common for both ground and instant coffee production, i.e. the roasting process, whilst SCGs are exclusive to the instant coffee process.

2.2. Inventory data and assumptions

The inventory data for ground and instant coffee production are presented in Tables 4–6 and Table S1 in the Supplementary information (SI). The data have been obtained from multiple sources, including Aspen Plus simulations, supporting calculations, open literature and the Ecoinvent database V3.3 [36]; for the details on the latter, see Table S2 in the SI. The life cycle stages shown in Fig. 1, data sources and corresponding assumptions are described in detail over the next sections.

2.2.1. Green coffee beans cultivation and transportation

Green coffee beans are sourced from the countries listed in Table 2 to reflect current coffee imports to the UK. The five-year (2014–2019) average data for imports have been sourced from the European Coffee Federation [13]. The arabica and robusta varieties of coffee are the only ones considered here because they represent the two main varieties traded internationally [37]. Whilst coffee imports by country commonly comprise a mix of these two varieties, it has not been possible to determine their composition due to a lack of data. Instead, it is assumed that the coffee imported from Brazil, Colombia and Honduras is arabica,

Table 2
Sources and types of coffee considered in this study.

Country of origin	Imports distribution (%)	Coffee variety[a]	Transport-domestic[b] (km)	Transport-international[c] (km)
Brazil	23	Arabica	200	7,000
Vietnam	16.2	Robusta	200	16,000
Colombia	8.6	Arabica	200	7,900
Honduras	4.3	Arabica	200	8,600
Other	47.9	Mix[d]	200	10,000

[a] According to the European Coffee Federation [13].
[b] Transportation assumed by truck.
[c] Transportation assumed by ship.
[d] 67% arabica and 33% robusta [13].

and the import from Vietnam is robusta [13]. A 67% arabica and 33% robusta mix is assumed for the other countries [13]. Such distribution is based on the UK's coffee imports data [13]. Life cycle inventory data for the cultivation and processing of green coffee beans (e.g. harvesting, cherry processing, etc.) have been sourced from Ecoinvent [36]. These data also include the impacts of land-use change.

The quantity of green beans required for producing ground and instant coffee varies according to the roasting level due to different yields [38,39] and production losses during sieving, grinding and other operations [40]. The corresponding green coffee beans-to-coffee product ratios estimated here based on mass balance calculations are presented in Table 3 and the specific quantities of green coffee beans are given in Table 4 and Table 5 for the ground and instant coffee, respectively. As can be observed in Table 3, the darker the roasting level, the more green coffee beans are required due to higher coffee losses during the roasting process.

Green coffee beans are first transported from their country of origin to the UK or the EU by ship. Based on the European Coffee Association, 79% of green coffee beans are shipped directly to the UK and the remaining 21% are sent to the EU for processing prior to being transported to the UK [13], assuming a combination of truck and ship transport. Once in the UK, it is assumed that the green coffee beans are transported from port to factory by truck. The assumed international and domestic distances are presented in Table 2.

The costs of green coffee bean production have been estimated based on the five-year (2014–2019) average prices sourced from the International Coffee Organization [3]. The associated transportation costs from the country of origin to the coffee factory have been estimated based on fuel consumption, i.e. heavy fuel oil for the ships and diesel consumption for the trucks. These are summarised in Table 4 and Table 5 for the two coffee products.

2.2.2. Ground and instant coffee manufacturing

As can be observed in Fig. 1, both ground and instant coffee production have several steps in common; these are shown in more detail in Fig. 2 and Fig. 3. In both cases, the green coffee beans are sieved to eliminate undesirable objects, such as gravel and wood, prior to being roasted. The clean green coffee beans are sent to the roaster, which is set at different temperatures depending on the desired degree of roasting. The roasted beans are cooled in a tray using ambient air and ground in a grinder to the desired particle size. The ground coffee is then packaged to produce the final ground coffee product. In the case of instant coffee, the ground coffee undergoes the extraction, evaporation and drying stages to produce a dry solid product which is then packed, ready for distribution. Two main drying processes are used – freeze and spray drying [25] – the former is assumed in the base case and the latter is considered in a sensitivity analysis.

These production processes have been simulated using Aspen Plus to obtain the necessary inventory data for LCA and LCC. Specifically, the modelling provided the data on utilities consumption (heat, electricity, refrigerants, water) and wastewater generated. The process has been modelled as a semi-continuous process in accordance with industrial practice [41]. A daily production of 100 tonnes of ground or instant coffee over 8000 h per year (333.3 days per year) is assumed [42]. Green coffee beans, roasted coffee and extracted coffee have been simulated as non-conventional solids, i.e. user defined, using physical properties sourced from Cardoso et al. [43] and Telis-Romero et al. [44], which are shown in Table S3.

The losses of green coffee beans have also been considered for both ground and instant coffee production [40,45]. A breakdown of these losses by process stage and roasting level is presented in Table S4.

Ground and instant coffee have been assumed to be produced in the UK (79.5%), Germany (5.4%), Italy (4.6%), France (2%), Netherlands (2%), Spain (1.4%) and other countries in the EU (5.1%), in line with the annual amount of coffee consumed in the UK [13,14]. The manufacturing process for both ground and instant coffee is assumed the same for all countries. However, the electricity mix for each of the abovementioned countries has been considered, sourced from Eurostat [46]; for details, see Table S5. The European mix is assumed for the other EU countries. The costs of utilities used in coffee production have been estimated using heat and electricity costs for non-household rates [47,48] while wastewater treatment costs have been sourced from United Utilities [49]. The process simulations of the manufacturing of ground and instant coffee are described in turn in the next sections.

2.2.2.1. Screening (ground and instant coffee).

Harvested green coffee beans usually contain around 2.2% of solid contaminants, such as sand, gravel and small pieces of wood [40]. To remove them, the green coffee beans are passed through a sieve as shown in Fig. 2 and Fig. 3.

2.2.2.2. Coffee roasting and grinding (ground and instant coffee).

The coffee is roasted in a batch roaster with hot air recirculation, as the latter is used commonly in practice to increase the energy efficiency [24]. The Aspen flowsheet of the coffee roasting process simulation is shown in Fig. 4. As can be observed, green coffee beans (GREEN: stream) are fed to the roaster (HEX: HeatX model) at room temperature (20 °C) and mixed with hot air (455 °C) [45], previously fed to the furnace at 20 °C. The furnace efficiency is assumed to be 90% and the pressure drop inside the roaster to be 1 kPa [50]. As previously mentioned, the beans are roasted at 192.5 °C for a light roast, 215 °C for a medium roast, 227.5 °C for a medium-dark roast and at 245 °C for a dark roast [6].

During the roasting process, the silverskin or chaff is removed from the green beans along with other impurities, such as carbon monoxide, methane and ash. The silverskin and ash are mixed with the hot air that is leaving the roaster and must be removed because part of this air is recycled back to the roaster to reduce energy consumption (SPLITER2: FSplit model). The removal of the impurities, which are assumed to be incinerated, has been simulated using a splitter (SPLITER1: FSplit model) and a mixer model (MIXER1: Mixer model). For the cyclone an SSplit model has been used to calculate the mass balance of the process (energy consumption is explained below). A trial and error method has been used to find the optimum splitter ratio for the recycled stream, where the cyclone's air outlet temperature is similar to the theoretical temperature for each roasting level (e.g. 215 °C for medium roast). The rest of the treated air is sent to an afterburner or thermal oxidiser (AFTERBUR: Heater model) operating at 760 °C [54], for treatment prior to be emitted to air (STACK: Stream). The roasting emissions considered here have been sourced from the US EPA [45]. At the end of the roasting process, the roasted beans are cooled down to 30 °C in a cooling tray (COOLING: HeatX model) using ambient air and ground in a grinder to the desired particle size. In the case of ground coffee, the product is sent to packaging, described in Section 2.2.3. For instant coffee, the ground coffee is sent to an extraction processing stage described in the next section.

The energy consumption of the cyclone and the grinding operations have been calculated separately, as the corresponding models in Aspen Plus require data on particle size distribution and work index for roasted coffee (grinding operation only) that has been unavailable at the time of the study. The energy requirements of the cyclone (SSplit model) have been estimated following the methodology presented in Pantaleo et al. [55] and using the following equation:

Table 3
Green coffee beans-to-coffee product ratio (weight by weight).

Roasting level	Green coffee beans-to-coffee product ratio	
	Ground coffee	*Instant coffee*
Light	1.16	2.32
Medium	1.21	2.43
Medium-dark	1.24	2.50
Dark	1.31	2.63

Table 4
Inventory and cost for different life cycle stages of ground coffee production (per tonne of coffee).

Life cycle stage[a]	Input	Unit (for quantity)	Roasting level								Data source	
			Light		Medium		Medium-dark		Dark			
			Quantity	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost*
Green coffee beans cultivation	Green coffee beans	kg	1159	1898	1214	1988	1245	2038	1308	2142	Own calc.[b]	[3]
Green coffee beans transportation	Heavy fuel oil	l	32.1	38.8	33.7	40.7	34.6	41.7	36.2	43.8	[36]	[51]
	Diesel	l	29.9	14.1	31.3	14.8	32.1	15.7	33.7	15.9	[36]	[51]
Packaging[c]	Aluminium	kg	6.1	7.9	6.1	7.9	6.1	7.9	6.1	7.9	Own meas., [16]	[52]
	PET[d]	kg	5.1	3.4	5.1	3.4	5.1	3.4	5.1	3.4	Own meas., [16]	[52]
	Polyethylene	kg	30.1	19.2	30.1	19.2	30.1	19.2	30.1	19.2	Own meas., [16]	[52]
	Polypropylene	kg	3.3	4.2	3.3	4.2	3.3	4.2	3.3	4.2	Own meas.	[52]

[a] For the manufacturing stage (energy data), see Table S1 in the SI.
[b] Based on data in Table 3.
[c] See Section 2.2.3 and Table 3 for further packaging details.
[d] Polyethylene terephthalate.

Table 5
Inventory and cost data for different life cycle stages of instant coffee production (per tonne of coffee).

Life cycle stage	Input	Unit (for quantity)	Roasting level								Data source	
			Light		Medium		Medium-dark		Dark			
			Quantity	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost*
Green coffee beans cultivation	Green coffee beans	kg	2318	3795.3	2432	3981.9	2495	4085.1	2625	4297.9	Own calc.[a]	[3]
Green coffee beans transportation	Heavy fuel oil	l	64.3	77.7	67.4	81.5	69.2	83.6	72.8	88.0	[36]	[51]
	Diesel	l	59.8	28.2	62.7	29.6	64.3	30.4	67.7	32.0	[36]	[51]
Coffee manufacturing[b]	Ammonia	kg	0.31	1.88	0.31	1.88	0.31	1.88	0.31	1.88	Own sim.	[53]
	Carbon dioxide	kg	0.30	0.22	0.30	0.22	0.30	0.22	0.30	0.22	Own sim.	[53]
Packaging[c]	Aluminium foil	kg	1.6	2.1	1.6	2.1	1.6	2.1	1.6	2.1	Own meas.	[52]
	Polypropylene	kg	118.5	152.1	118.5	152.1	118.5	152.1	118.5	152.1	Own meas.	[52]
	Paper	kg	25.4	0.9	25.4	0.9	25.4	0.9	25.4	0.9	Own meas.	[52]
	Glass	kg	2058	39.1	2058	39.1	2058	39.1	2058	39.1	Own meas.	[52]

[a] Based on data in Table 3.
[b] For energy data, see Table S1 in the SI.
[c] See Section 2.2.3 and Table 3 for further packaging details.

$$E_{cyclone} = P{\cdot}t{\cdot}R \quad (1)$$

where:

E energy consumption of cyclone (kJ)
P power rating of cyclone (assumed at 44 kW [55])
t roasting time (assumed at 600 s, 680 s, 785 s and 785 s for light, medium, medium-dark and dark roast, respectively)
R green coffee beans-to-coffee product ratio (see Table 3).

For the grinding operation, the average energy consumption of various grinders reported in Piccinno et al. [56], equal to 16 kWh/t, has been assumed in this study (see Table S1).

2.2.2.3. Extraction and evaporation (instant coffee). The extraction process comprises six extraction columns [57]. It has not been possible to simulate the extraction process in Aspen Plus due to a lack of coffee solubility data; instead, a pseudo-extraction simulation based on a mass balance around the columns has been used. The heat capacity and density of the extracted coffee required as an input for the simulation has been sourced from Telis-Romero et al. [46]. The properties are shown in and the resulting flowsheet is presented in Fig. 5. As can be observed, heat exchangers, splitters and mixers have been used to model the extraction process, assuming cross flow [25,57]. The ground coffee is distributed equally between the extraction columns. Coffee extraction takes place at two different temperatures due to the complex composition of coffee beans and to avoid the decomposition of some of its key components [41,58]. The first four columns or 'hot cells' operate at 140 °C and are used to extract carbohydrates and lipids present in coffee. The last two columns (known as 'cold cells') operate at 100 °C and are used to extract the compounds that give coffee its characteristic aroma, such as phenolic compounds, aliphatic acids and chlorogenic acids [41]. Coffee is extracted using water at 175 °C and 793 kPa, at a water-to-ground coffee ratio of 3:1 [25]. The column volume-to-coffee weight ratio is assumed to be 2.63 [25]. The extraction yield increases progressively as coffee passes through each column, starting at 2.4% and increasing to 10%, 18.75%, 25%, 35% and finally reaching 50%, which is the assumed extraction yield of the process [41]. The amount of wastewater and SCGs discharged from the columns has been calculated as follows [25]:

$$w = 0.844[V - ((X - x)/1.2)] \quad (2)$$

where:

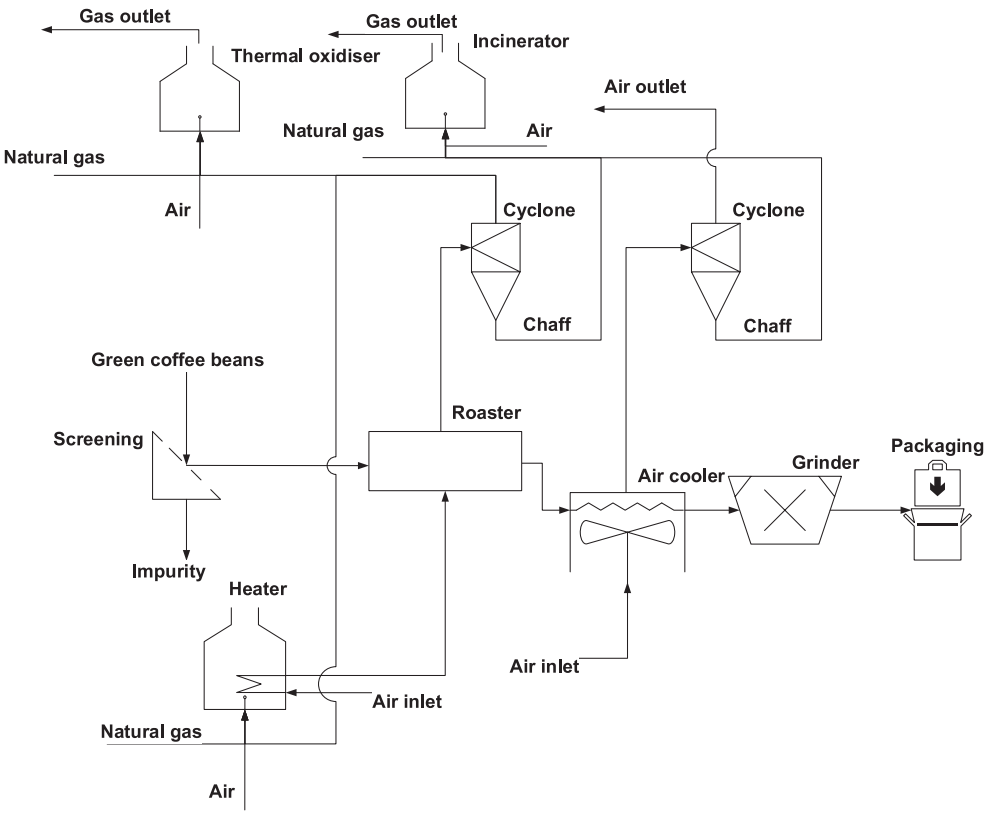


Fig. 2. Process flow diagram of ground coffee production.

w	amount of water and the SCGs discharged from the columns (kg)
V	volume of percolation column (l)
X	amount of roasted coffee fed to the column (kg)
x	amount of moisture in the feed (assumed at 3.5% of the feed, kg).
1.2	inherent particle density of SCGs (kg/l)
0.844	density of water at 175 °C (kg/l).

The extracted coffee, containing 75% of water, enters an evaporator (EVAP: HeatX model) at 100 °C and leaves at 70 °C with the water content reduced to 42% [58]. In the simulation, the extracted coffee stream is used to pre-heat the incoming water (from 20 °C to 88 °C) used in the extraction unit, as shown in Fig. 5. This is to reduce the heat consumption used in the process. The SCGs, which is the waste resulting from this process stage, are assumed to leave the system untreated to allow for a fair comparison with ground coffee production. Nevertheless, the treatment of SCGs via incineration (common treatment of SCGs [59,60]) is explored in a sensitivity analysis in Section 3.3.3.

2.2.2.4. Freeze drying (instant coffee). The freeze-drying process occurs in the freezing room, where the concentrated extracted coffee is frozen and ground, followed by a sublimation chamber where the moisture is removed from the frozen granules to obtain a dry solid product. The process has been simulated using a three-level refrigeration cycle: a two-level NH_3 cascade and a CO_2 cycle [61] as shown in Fig. 6. First, the concentrated extracted coffee is cooled down from 70 °C to 30 °C in a heat exchanger (HEX1: HeatX model) using cooling water. This is then fed to a refrigerator to reduce the temperature in three steps, starting at −5 °C (NH_3 upper refrigeration cycle), followed by −28 °C (NH_3 lower refrigeration cycle) and finally to −50 °C (CO_2 refrigeration cycle). The frozen coffee is then granulated and sent to a sublimation chamber (SUBLIMAT: Sep), which is operated under vacuum (631 Pa) until the moisture content is reduced to 4% wt. [41]. The energy consumption of the coffee granulation step has been calculated separately, assuming 16 kWh/t [56] in the same way as for coffee grinding, as both unit operations have the same principle of reducing the size of solid particles.

The amounts of refrigerants used in the refrigeration cycle for both CO_2 and NH_3 have been obtained from the Aspen modelling, assuming a 15% annual leakage [62]. The refrigerant costs have been estimated using the refrigerant average costs available on the market [53].

2.2.2.5. Coffee pods and instant-coffee mix. Coffee pods contain ground coffee and hence the manufacturing process of coffee is the same as shown in Fig. 2, with the inventory provided in Table 4 and Table 6.

The production process of the instant-coffee mix is similar to that for instant coffee, except that it has additional ingredients. According to own market survey, the main ingredients in the instant-coffee mix are instant coffee (11%), milk powder (48%) and sugar (41%). The LCA and LCC data for freeze-dried instant coffee presented previously have been used, while the LCA data for milk powder and sugar have been sourced from Sieti [63] and the Ecoinvent database [36], respectively. The LCC data for these two products have been estimated based on data from the Office for National Statistics [64], Defra [65] and WRAP [52]; for the inventory data, see Table S6 and S7.

2.2.3. Packaging

Different types of packaging are considered for different types coffee, based on the products available on the UK market. The packaging-related data are summarised in Table 4 and Table 5 for the ground and instant coffee, respectively, with further details provided Table S7 for all types of coffee.

Ground coffee is assumed to be packed in laminated pouches (227 g/pack), consisting of polyethylene (PE), aluminium, polyethylene terephthalate (PET) and a CO_2-release valve made of polypropylene (PP) [16]. The valve allows the CO_2, still being released after the roasting process, to escape and preserve the quality of the coffee [66]. The CO_2 released from the packaged coffee has not been taken into account due to a lack of data.

Instant coffee is assumed to be packed in a glass jar (200 g/pack)

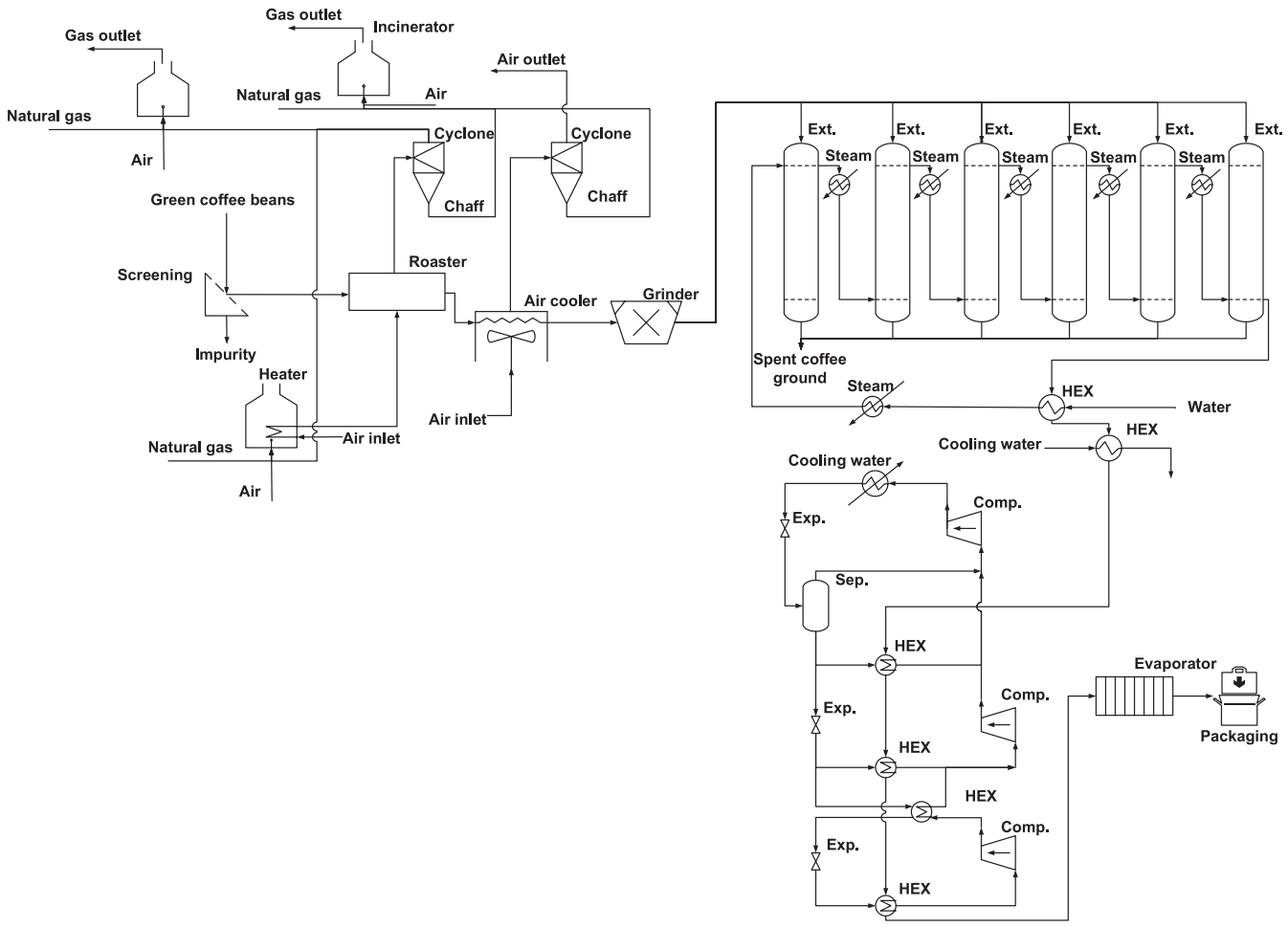


Fig. 3. Process flow diagram of instant coffee production. [Comp. = compressor, Cyc. = cyclone, Ext. = extraction column, Exp. = expansion valve, HEX = heat exchanger, Sep. = separator].

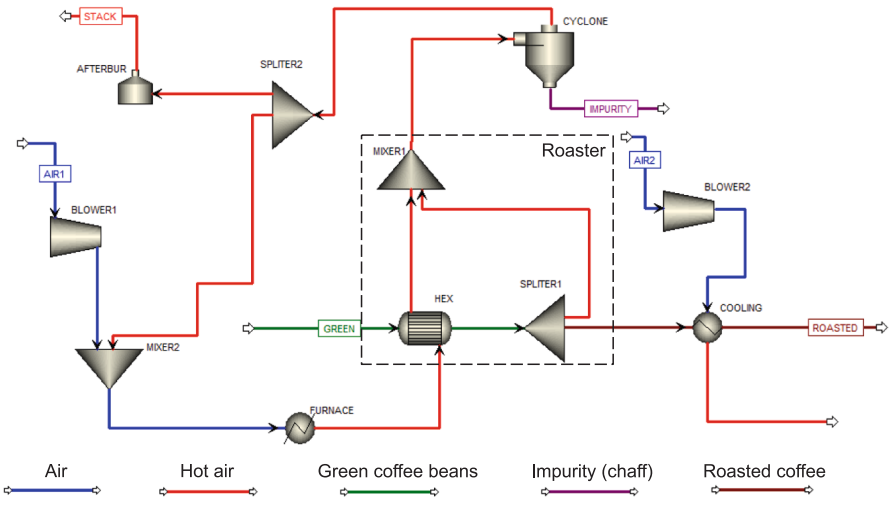


Fig. 4. Process flowsheet of coffee roasting with hot air recirculation. [STACK: emissions to air; HEX: heat exchanger; COOLING: cooling tray].

sealed with aluminium foil and closed with a PP lid; paper labelling is used for product information. Coffee pods (5.7 g/pod) are made from aluminium, which are packed in paper boxes (10 pods/box). Finally, the instant-coffee mix is packaged in aluminium or plastic sachets, which are packed in a paper box (6 sachets/box). For the packaging data, see Table S7.

It is assumed that the packaging is made from a mix of virgin and recycled materials (Table S8). The system has been credited for displacing the virgin materials, but taking into account the impacts of recycling. The recycling rates are 52% for aluminium, 79% for paper,

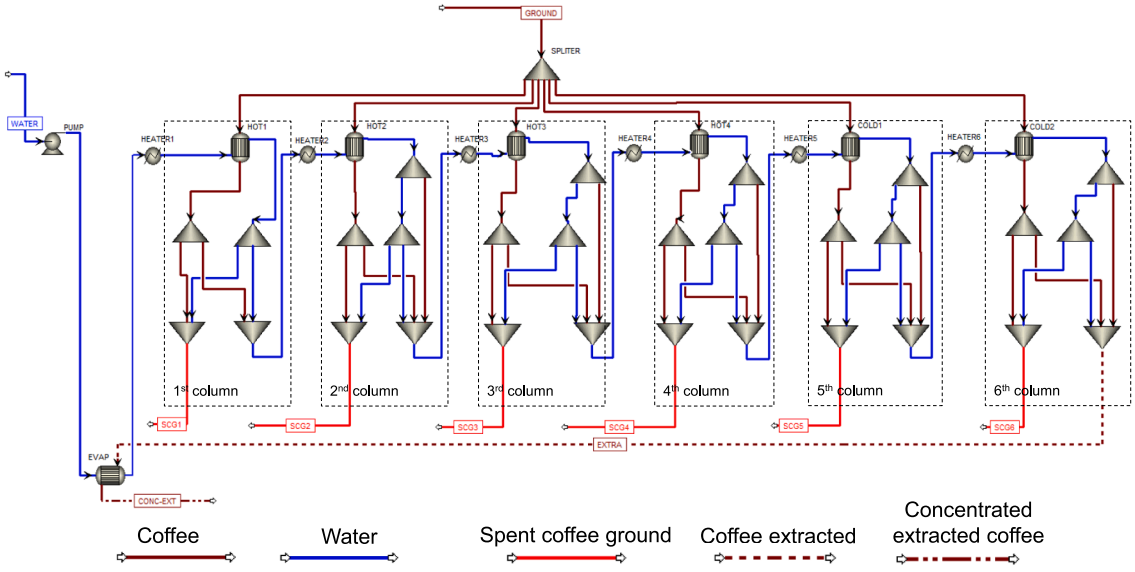


Fig. 5. Process flowsheet for instant coffee extraction and evaporation process stages. [Ground: ground coffee; SCG: spent coffee grounds; EVAP: evaporator; EXTRA: extracted coffee; CONC-EXT: concentrated extracted coffee; HOT: hot cell; COLD: cold cell].

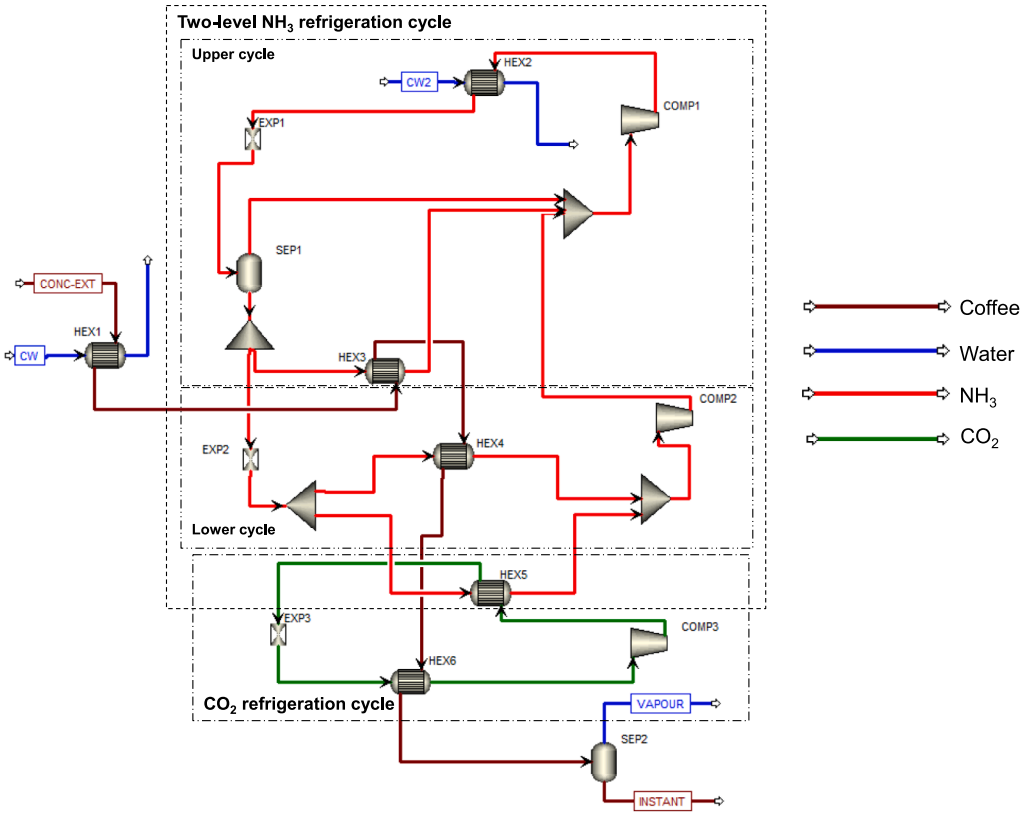


Fig. 6. Process flowsheet for the instant coffee freeze-drying process stage. [CW: cooling water; CONC-EXT: concentrated extracted coffee; INSTANT: instant coffee; COMP: compressor; HEX: heat exchanger; EXP: expansion valve; VAPOUR: removed moisture].

67.6% for glass and 46.2% for plastics (except PP which is currently not recycled for food packaging) [67,68]. The costs of packaging have been estimated based on data published by WRAP [52] considering the costs of both virgin and recycled materials. Further detail on the estimation of packaging costs can be found in Section S1 and Tables S7 and S8 in the SI.

2.3. Environmental impact assessment and costing

The ReCiPe 2016 V1.1 impact assessment method has been used to estimate the environmental impacts assuming a hierarchist perspective [69]. The environmental impacts are assessed at the midpoint level. The following 18 impacts have been estimated: climate change (CC); fossil depletion (FD); freshwater eutrophication (FE); freshwater ecotoxicity

Table 6
Waste and emissions from coffee manufacturing and costs of treatment (per tonne of coffee).

Coffee product	Waste/Emissions	Unit (for quantity)	Roasting level								Data source	
			Light		Medium		Medium-dark		Dark			
			Quantity	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost (£)*	*Quantity*	*Cost*
Ground coffee	PM_{10}[a]	g	109	–	114	–	117	–	123	–	[45]	–
	Volatile organic[a] aacompounds[a]	g	25	–	26	–	27	–	28	–	[45]	–
	Carbon monoxide[a]	g	289	–	303	–	311	–	326	–	[45]	–
	Carbon dioxide[a]	kg	279	–	292	–	299	–	314	–	[45]	–
	Methane[a]	g	79	–	83	–	85	–	89	–	[45]	–
	Silverskin[b]	kg	45.9	0	47.1	0	47.8	0	49.2	0	Own calc.	–
Instant coffee	Wastewater	m^3	0.1	0.13	0.1	0.13	0.1	0.13	0.1	0.13	Own sim. [49]	[49]
	PM_{10}[a]	g	218	–	228	–	234	–	246	–	[45]	–
	Volatile organic compounds[a]	g	49	–	52	–	53	–	56	–	[45]	–
	Carbon monoxide[a]	g	578	–	607	–	622	–	655	–	[45]	–
	Carbon dioxide[a]	kg	557	–	585	–	600	–	631	–	[45]	–
	Methane[a]	g	158	–	165	–	170	–	179	–	[45]	–
	Silverskin[b]	kg	97.4	0	102.1	0	104.8	0	110.3	0	Own calc.	–
	Spent coffee grounds[c]	kg	4026	–	4026	–	4026	–	4026	–	Own calc.	–

[a] Emitted to air and not treated; hence, the costs are not applicable.
[b] No external fuel is required for the incineration of silverskin.
[c] 76.1% moisture content. Outside the scope of the study.

(FET); human toxicity - cancer (HTC); human toxicity - non-cancer (HTNC); ionising radiation (IR); land use (LU); metal depletion (MD); marine eutrophication (ME); marine ecotoxicity (MET); ozone depletion (OD); particulate matter formation (PMF); photochemical ozone formation - ecosystem (POFE); photochemical ozone formation - human health (POFH); terrestrial acidification (TA); terrestrial ecotoxicity (TET); and water depletion (WD). Primary energy demand (PED) is also considered and has been determined using the method available in GaBi [32]. The biogenic carbon and the primary energy demand of biomass (from the Sun) are excluded in accordance with the ISO 14047 standard [70,71].

The LCC have been estimated according the following equation:

$$LCC = C_{RM} + C_T + C_P + C_U \quad (3)$$

where:

LCC life cycle costs of coffee (£/t)
C_{RM} costs of raw materials (£/t)
C_T transportation costs (£/t)
C_P cost of packaging (£/t)
C_U cost of utility (£/t).

All costs are based on 2018 prices (except for green coffee beans which are based on a five-year average). The costs of infrastructure, labour and maintenance are excluded for consistency with LCA. The tax related to utilities cost such as heat, electricity and water are included.

For the sectoral analysis, the total LCC have been estimated based on the sales volume of coffee reported by the British Coffee Association [4]. The annual amounts of coffee sold are 38 kt of instant, 19 kt of ground, 13 kt of pods and 3 kt of instant-coffee mixes. As the roasting level data are not available for all of these products, four possible scenarios have been assumed, assuming different consumer preferences:

- Scenario 1: equal distribution (25% for all roasting levels)
- Scenario 2: dark roast is preferred (40% dark roast, 30% medium-dark roast, 20% medium roast, 10% light roast)
- Scenario 3: light roast is preferred (40% light roast, 30% medium roast, 20% medium-dark roast, 10% dark roast)
- Scenario 4: medium/medium-dark roast is preferred (40% medium roast, 40% medium-dark roast, 10% light roast, 10% dark roast).

3. Results and discussion

This section presents first the results of process modelling for the eight coffee options considered, which were used as inventory data for LCA and LCC. This is followed in Section 3.2 by the environmental impacts and costs of coffee manufacturing assuming a gate-to-gate system boundary. The equivalent results for the cradle-to-gate boundary are presented in Section 3.3. For both system boundaries, the results are expressed as per tonne of ground or instant coffee. Finally, the results are scaled-up in Section 3.4 to estimate the total annual impacts and costs of coffee production at the sectoral level.

3.1. Process modelling of ground and instant coffee production

The heat and electricity consumption for the ground and instant coffee manufacturing processes estimated in this study are presented in Table S1 (as these data are used as inputs into the LCA and LCC, they are presented in the Methods rather than the Results section, but are discussed here). The majority of these values have been obtained through the process simulations carried out in Aspen Plus, with the exception of a few unit operations, which have been considered separately due to software limitations, as explained in Section 2.2.2.

As can be observed in Table S1, the ground coffee production process consumes between 702.7 and 1121.4 MJ of heat, with the coffee roasting stage requiring 66–67% of the total heat to increase the air temperature. This is followed by the afterburner, which consumes the remaining 33–34% of the heat to treat air emissions. The instant coffee production process requires between 8672.9 and 9535.3 MJ of heat, which is 8.5–12 times higher than for ground coffee. This is mostly due to the heat utilities associated with operating the sublimation chamber in the freeze-drying stage (42.5–47%) and water heating for coffee extraction (33–37%). The roasting process contributes 11–16% while the afterburner is responsible for the remaining 5–8%. It is important to mention that the heat consumption during the roasting and afterburner stages will depend on the amount of green coffee beans fed into the process, which varies with the roasting levels. The heat consumption for the extraction, evaporation and freeze-drying stages, on the other hand,

will remain constant for all roasting levels because the same amount of coffee is processed to produce instant coffee.

In addition to the heat, the production of ground coffee consumes 77.8–95.8 MJ of electricity. Coffee grinding and the cyclone are the main consumers, contributing 46–57% and 39–50% to the total electricity demand, respectively; the cooling stage uses only 4%. Similar to the trends observed for heat consumption, instant coffee requires 7.6–8.5 times more electricity (662.8–700.2 MJ/t) than ground coffee. This is largely due to the electricity used for freeze-drying to operate the compressors in the refrigeration cycle (48–51%). The granulating (frozen extracted coffee) and grinding (roasted coffee) processes account together for 37–39% of the total consumption, making it a second hotspot, followed by the cyclone operation (9–14%). The cooling accounts for <1% of the total.

The instant coffee production process consumes more energy overall because the green coffee beans- to-coffee product ratio is higher (see Table 3). In other words, more green beans are needed to produce the same amount of product. In both cases, the overall process energy demand increases with the roasting level as the roasting temperature increases, as well as the amount of green beans required due to greater losses in the process.

3.2. Gate-to-gate system boundary

The environmental impacts are discussed in the next section, starting with climate change as one of the main policy drivers and followed by the other environmental impacts. A comparison with other LCA studies is presented in Section 3.2.1.1, while the LCC results are discussed in Section 3.2.2. Finally, the results of sensitivity analyses are presented in Section 3.2.3.

3.2.1. Environmental impacts

As can be seen in Fig. 7 and Table S9 in the SI, the CC impact is estimated at 400–453 kg CO_2 eq./t for ground coffee and 1951–2056 kg CO_2 eq./t for instant coffee. The main reasons for a five-times higher impact of instant coffee are its glass packaging and the higher energy requirements in the coffee production process (as discussed in Section 3.1).

For the production of ground coffee, the main hotspot is the CO_2 emission from coffee roasting, which is responsible for 70% of the total CC at all roasting levels. This is followed by the manufacture of the laminated pouch used for packaging which causes 19–21% of the impact, mainly due to PE, with some contribution from aluminium (5–6%). Utilities contribute 8–10% to the total CC impact, mainly due to heat consumption, while electricity adds only 2%.

In the case of instant coffee, the manufacture of the packaging is the single highest contributor to CC (49–51%), mostly due to the glass jar and polypropylene lid. The roasting emissions are responsible for 29–31% of the total impact, followed by heat with 15–16%, with the latter mostly due to the energy requirements for freeze drying and coffee extraction. Like ground coffee, electricity consumption contributes causes only 3% of CC.

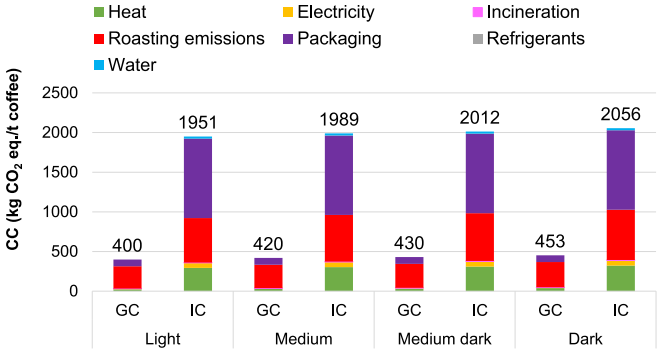


Fig. 7. Gate-to-gate climate change impact of ground and instant coffee production for different roasting levels (see Table S9 for further details).

As can also be inferred from Fig. 7, the roasting level influences the impact of both ground and instant coffee. For example, the CC of a dark roast ground and instant coffee is 13% and 5% higher than the equivalent light roasted product. This is due to the higher roasting temperatures required for the darker roasts and the corresponding energy requirements (see Table S1).

The other impacts of instant coffee are 2–42 times higher than those of ground coffee (Fig. 8). This is due to a combination of factors including the use of glass jars for packaging and energy and water intensive processes involved in instant coffee production. Packaging is the dominant hotspot for most impact categories for both types of coffee. It contributes 59–96% to the impacts of ground coffee and 48–92% for instant coffee due to the use of PE and aluminium, and glass and PP, respectively. Heat consumption is a second hotspot in instant coffee production for FD, OD and PED, contributing around 20% to each. The incineration of the remaining ash and silverskin not removed by the cyclone after the roasting process represents the main hotspot for POFE (95–96%), POFH (78–79%) and OD (48%) for ground coffee, and POFE (79–80%) and POFH (95%) for instant coffee. This is largely due to the combustion emissions released to the environment. WD is another hotspot, causing 89% of the total impact from instant coffee production. This is because the production of instant coffee requires the brewing of coffee (see Table 4). Electricity consumption is the second hotspot for IR (27–37%) and LU (11–38%) for both types of coffee. The former is related to nuclear power and the latter to the land used to cultivate biomass used for electricity generation. For further details, see Table S9 in the SI.

The roasting levels affect mainly FD, FE, HTNC, POFE and POFH but the effect is relatively small. For example, the FD of dark roast ground and instant coffee is 10% and 2% higher, respectively, than that of the light roast (Fig. 8).

3.2.1.1. Comparison with other studies. The results obtained in this study have been compared with other similar studies reported in the literature [15,18,19]. The comparison has only been possible for CC as other studies used different impact assessment methods for the other impacts. Some of the original results have been recalculated here using the functional unit of this study for comparison purposes as they varied across studies.

As shown in Fig. S1 in the SI, the gate-to-gate CC impact of ground coffee reported in the literature ranges between 0.38 and 0.62 t CO_2 eq./t and 0.19–0.37 t CO_2 eq./t when packaging is excluded (to make results comparable as packaging type varied or was not considered in different studies). These values are in agreement with the ranges estimated in this study of 0.40–0.45 t CO_2 eq./t and 0.31–0.37 t CO_2 eq./t, respectively (excluding the packaging). The small differences in the results could be related to different energy mixes and GHG characterisation factors considered.

Only one study of instant coffee is available, which considered a dark roast produced in Japan [19]. The results in that and the current study are close (1.33 t CO_2 eq./t and 0.95–1.05 t CO_2 eq./t, respectively, excluding packaging). The slight differences are due to different production processes (spray- vs freeze-dried) and differences in the energy mixes between the UK and Japan.

3.2.2. Life cycle costing

The results in Fig. 9 reveal that the LCC of ground coffee are estimated at £43-48/t. The packaging is the main hotspot with £34.7/t, contributing 73–80% to the total costs. Among the materials, PE has the highest contribution (55%) to the packaging costs, followed by aluminium (22%), PP (12%) and PET (10%). The LCC of instant coffee are around eight times higher (£375-384/t) than those of ground coffee

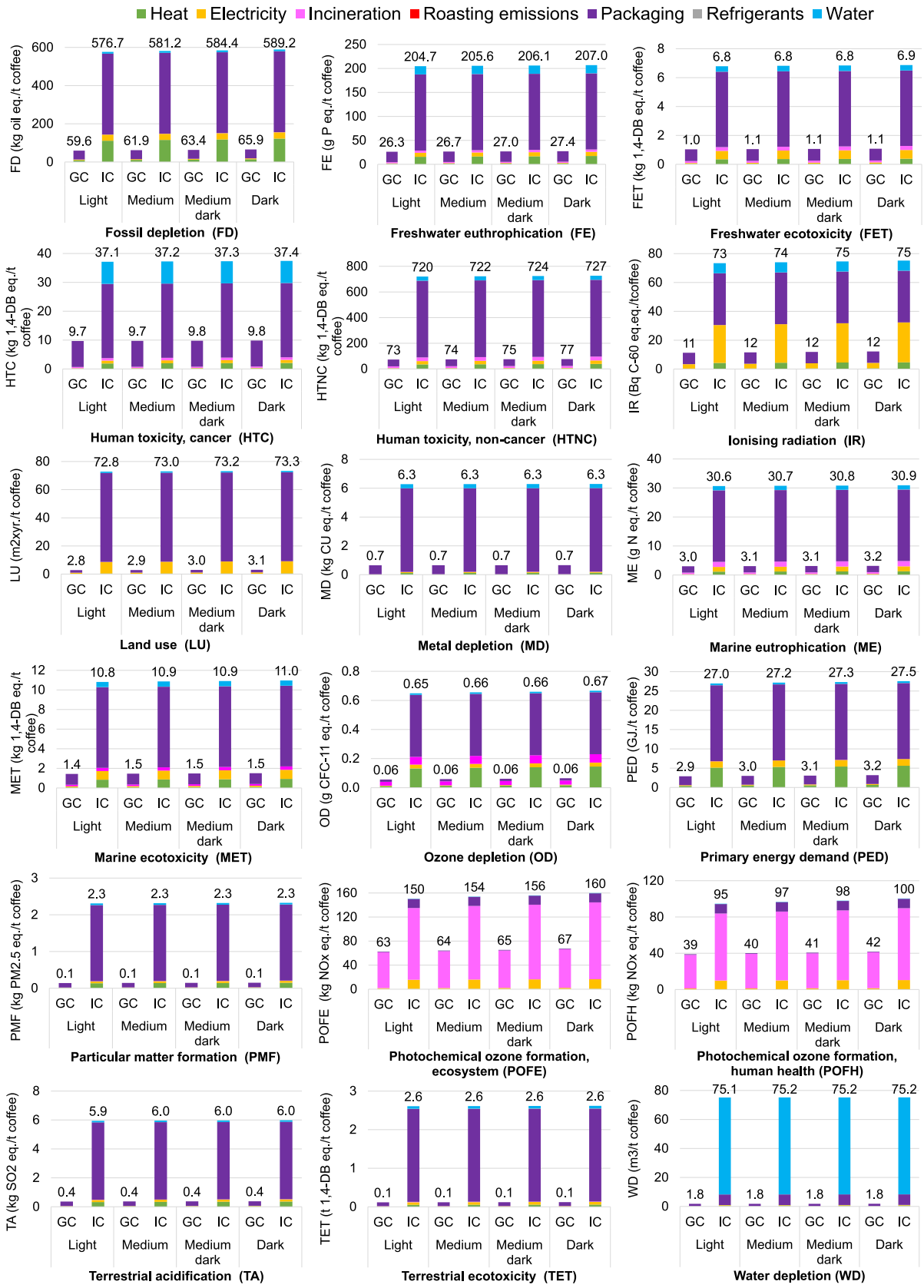


Fig. 8. Other gate-to-gate impacts of ground and instant coffee production for different roasting levels (see Table S9 for further details). [GC = ground coffee; IC = freeze-dried instant coffee].

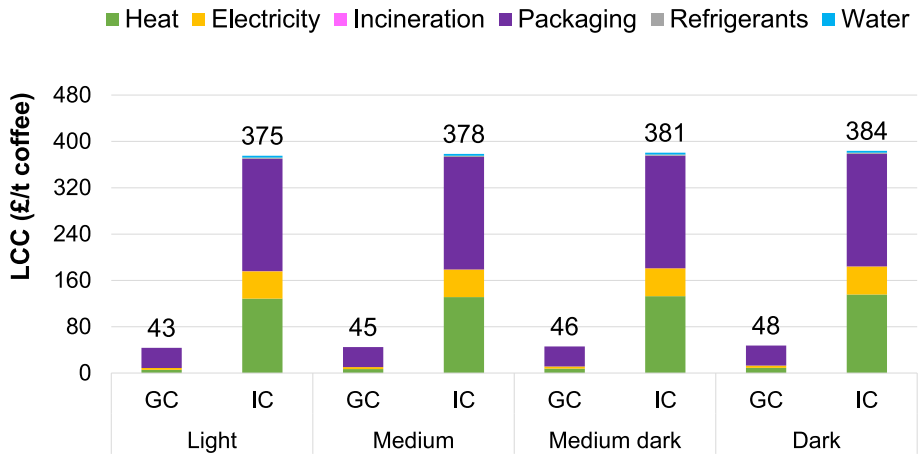


Fig. 9. Gate-to-gate life cycle costs (LCC) of ground (GC) and instant (IC) coffee production for different roasting levels (see Table S9 for further details).

because of the glass packaging, which is more expensive than laminated pouches and requires significantly more energy to produce. The glass packaging contributes 51–52% to the total LCC of instant coffee, with the remaining costs being due to the heat (34–35%) and electricity (13%) consumption in the coffee production process.

Similar to the climate change trends, the roasting levels have an effect on the LCC of both ground coffee and instant coffee, with a 12% and 2% difference between the light and dark roasts, respectively. The contributions from the roasting process are less pronounced in the instant coffee production as this type of coffee is subjected to other energy-intensive processes (extraction and freeze drying) in addition to roasting.

3.2.3. Sensitivity analysis

To address the main hotspots identified in the study (packaging, waste treatment and heat consumption, the latter for instant coffee), the following two parameters are considered in the sensitivity analysis: alternative packaging and energy recovery from silverskin. In addition, alternative coffee roasting and instant coffee production processes are also considered to assess the influence on the impacts and costs.

3.2.3.1. Alternative packaging. This section examines the environmental and economic implications of alternative packaging materials for both types of coffee based on alternative packaging available on the market. The packaging considered here is aluminium pods and C/LDPE (also known as aluminium/LDPE) pouch for ground coffee, and laminated pouch for instant coffee. The materials required for different types of packaging are provided in Table S7 in the SI.

The gate-to-gate impacts and costs of ground and instant coffee packed in alternative packaging are presented in Table S10 and the relative change compared to the base case in Table S11. The results for ground coffee indicate that the use of pods instead of laminated pouches would increase all environmental impacts (4–1477%), and in particular MD and HTC, which would be 25–26 and 21 times higher, respectively. This is because a pod requires 28 times more aluminium than a laminated pouch. Conversely, C/LDPE pouch would reduce all impacts by 1–29% compared to the laminated alternative as it requires 28% less LDPE. In the case of instant coffee, the laminated pouch could also reduce the impacts significantly (8–89%), particularly TET, TA and LU.

The costs show a similar trend to the environmental impacts. For example, if the packaging for ground coffee changes from laminated pouch to aluminium pods, the gate-to-gate costs would increase by 409–447%. On the other hand, the use of C/LDPE packaging would reduce the costs by 27–29%. For instant coffee, if glass jars are replaced with laminated packaging, the costs would be around 43–44% lower.

3.2.3.2. Energy recovery from silverskin. As discussed in Section 3.2.1, incineration of ash and silverskin is the main hotspot for POFE, POFH and OD for ground coffee, and POFE and POFH for instant coffee. Silverskin has a lower heating value (LHV) of 18.31 kJ/kg [72] and can be used as a fuel. Therefore, this section explores the environmental and economic impacts of coffee production if silverskin is used to produce heat used in the system and replacing an equivalent amount of natural gas. The amount of silverskin generated and incinerated is shown in Table S12, along with the amount of heat it produces.

The results in Table S13 and S14 reveal that using silverskin as a fuel would reduce all environmental impacts and costs of both ground and instant coffee production. This is particularly significant for POFE, POFH and OD of ground coffee which would be 65–96% lower, and for POFE and POFH of instant coffee, which would reduce by 78–80%. This is due to the heat from silveskin replacing the equivalent amount of heat from natural gas. In addition, the LCC of ground coffee are reduced by around 15% for the same reason, whereas only a 4% reduction can be observed for instant coffee because heat consumption is not a main hotspot and has a lower effect on the costs.

3.2.3.3. Roaster without hot air recirculation. Roasting is the most important step in coffee production as it gives its distinctive aroma and flavour [73]. It is also one of the hotspots, contributing up to 35% to the impacts and costs of ground coffee and 22% for instant coffee. Therefore, the 'single-pass' roasting method, i.e. without hot air recirculation, which is also commonly used in industry [38], is assessed here and compared to hot air recirculation assumed in the base case to determine the effect on the impacts and costs.

Aspen Plus was used to model the 'single pass' roasting method and the corresponding flowsheet is shown in Fig. S2 (for comparison with the hot air recirculation process, see Fig. 4). The heat and electricity consumption of the 'single pass' method are shown in Table S15, indicating that using the 'single pass' system increases heat consumption in the production of ground coffee by 146–167%, whereas no significant changes could be observed for electricity consumption (<3%). For the production of instant coffee, heat consumption increases by 24–40% and 1% for electricity consumption.

The increase in heat consumption leads to a significant increase in all impacts and LCC due to the additional heating fuel required for the process. The most affected impacts are FD, OD and PED which are now 21–44% higher compared to the base case (Table S16 and S17). Similarly, the costs increase by 19–36%.

3.2.3.4. Spray-dried instant coffee. This sensitivity analysis looks at the environmental impacts and costs associated with the production of spray- instead of freeze-dried instant coffee, which was also simulated in

Aspen Plus (Fig. S3).

The production of spray-dried instant coffee is similar to the freeze-drying process, except for the concentration of the extracted coffee and the drying method. The concentration of the extracted coffee is 50% at the evaporator outlet [58] and a spray-dryer is used to dry the concentrated extracted coffee. The inlet and outlet hot air temperatures for the spray-dryer are assumed to be 270 and 110 °C, respectively [74,75].

The energy consumption of the production of spray-dried instant coffee is shown in Table S18. It can be seen that the heat consumption (8754–9616 MJ/t) is quite similar to freeze-drying (<1% difference). The electricity consumption of the process, on the other hand, is 71.6–75.4% lower (162–198 MJ/t). This is because spray-drying does not required refrigeration, which is the main electricity consumer in the freeze-drying process.

The environmental impacts and LCC of spray-dried coffee in Tables S19 and S20 suggest that its environmental impacts are similar to freeze-dried coffee, with less than a 9% difference. The only exception is IR, which is 27–28% lower due to less electricity required. The LCC of spray dried-instant coffee are 6% lower than freeze-dried instant coffee for the same reason.

3.3. Cradle-to-gate system boundary

3.3.1. Environmental impacts

Considering the cradle-to-gate impacts, it can be seen in Fig. 10 and Table S21 that the production of ground coffee generates 7.0–7.9 t CO_2 eq./t and twice as much for the production of instant coffee (15.2–17 t CO_2 eq./t). As opposed to the results for the gate-to-gate boundary, packgaging is no longer the hotspot, now replaced by the cultivation of green coffee beans. The latter causes 84–85% and 91% of the total CC impact of the ground and instant coffee, respectively. This is mainly due to the field emissions from land-use change (56% and 52%) and from nitrous oxide emissions related to the use of fertilsers, fuels and treatment of waste (27% and 25%). The contribution of packaging is slightly higher for instant (6–7%) than for ground (3–4%) coffee. The contribution of the production process is relatively small (5–6%), as is of transport (1–3%), for both coffee types. When looking at the effect of different roasting levels, the dark roast has a 12–13% higher CC impact than the light-roast variety (Fig. 10).

The other environmental impacts of instant coffee are 2–2.3 times higher compared to ground coffee for all impact categories (Fig. 11 and Table S21). Similar to CC, the cultivation of green coffee beans is the main hotspot, contributing >87% and 75% to the impacts. This is due to the use of fertilisers, agricultural machinery and fuel consumption. Packaging is also significant for FD and PED of instant coffee, with a contribution of 11–13% to the impacts due to the impacts of glass and PP. Transport causes 5–11% of FD, IR, TET and PED of both types of coffee. The contribution of the production process of ground coffee to all other impacts is small (<5%), while that of instant coffee is even lower (<2%). Similar to CC, the other impacts of dark-roast coffee are 11–13% higher than the light-roast equivalents.

3.3.2. Comparison with other studies

The comparison of the climate change impact of ground coffee with the literature [15,18-21] can be found in Fig. S4. Comparisons for the other impacts are not possible due to different LCA methods used in the literature. Furthermore, comparison for instant coffee is not possible due to a lack of data in the literature. As can be seen in Fig. S4, the cradle-to-gate CC impact of ground coffee reported in most sources ranges from 5.2 to 8.2 t CO_2 eq./t [15,18-21]. This compares well with the values obtained in this study (7–7.9 t CO_2 eq./t). The only exception is the value of 1.7 t CO_2 eq./t reported by Hick [20], which is an outlier. The main differences observed across these studies are for coffee bean cultivation, which varied with regions, coffee yields and agricultural practices [24]. For example, Hassard et al. [19] considered coffee grown in Costa Rica and Guatemala (4.31 t CO_2 eq./t), Humbert et al. [15] focused on coffee in Brazil (4.71 t CO_2 eq./t) and Brommer et al. [18] sourced coffee from Tanzania (3.31 t CO_2 eq./t), whilst this study considered several countries (6.35–7.17 t CO_2 eq./t). Nevertheless, the results of this study are congruent with the literature as the cultivation of green coffee beans were also found to be the main hotspot across all the studies [15,18-21].

3.3.3. Life cycle costing

The LCC are estimated to be £1994-2249/t for ground and £4277-4802/t for instant coffee (Fig. 12). Like the environmental impacts, the majority of the costs are related to the cultivation of green beans, contributing 95% to the costs of ground and 89–90% to those of instant coffee. The pouch adds 2% for ground coffee (£34.7/t), whilst glass packaging for instant coffee accounts for 4–5% of total costs (£194/t). Transportation increases the total costs by a further 3% for both ground (£53-60/t) and instant coffee (£106-120/t). The remaining costs are due to the production process (£53-60/t for ground and £181-190/t for instant coffee). Dark roast has 12–13% higher LCC than light roasts for both ground and instant coffee.

As far as the authors are aware, there are no similar economic assessment studies of ground and instant coffee reported in the literature and for this reason it is not possible to compare these results with literature. However, for context, the average retail price of ground coffee in the UK was £16.6/kg (or £16,600/t) and that of instant coffee was £26.7/kg (£26,700/t), based on own survey for the same base year (2018). The difference with the costs estimated here is due to the labour, storage, maintenance and capital costs, as well as profits, which have not been considered in this study.

3.3.4. Sensitivity analysis: Treatment of spent coffee grounds

This section explores the effect on the environmental impacts and

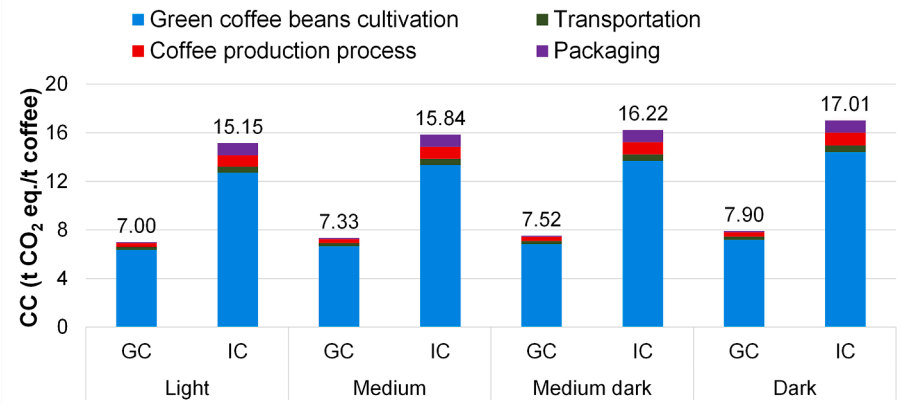


Fig. 10. Cradle-to-gate climate change impact of ground and instant coffee production for different roasting levels (see Table S21 for further details).

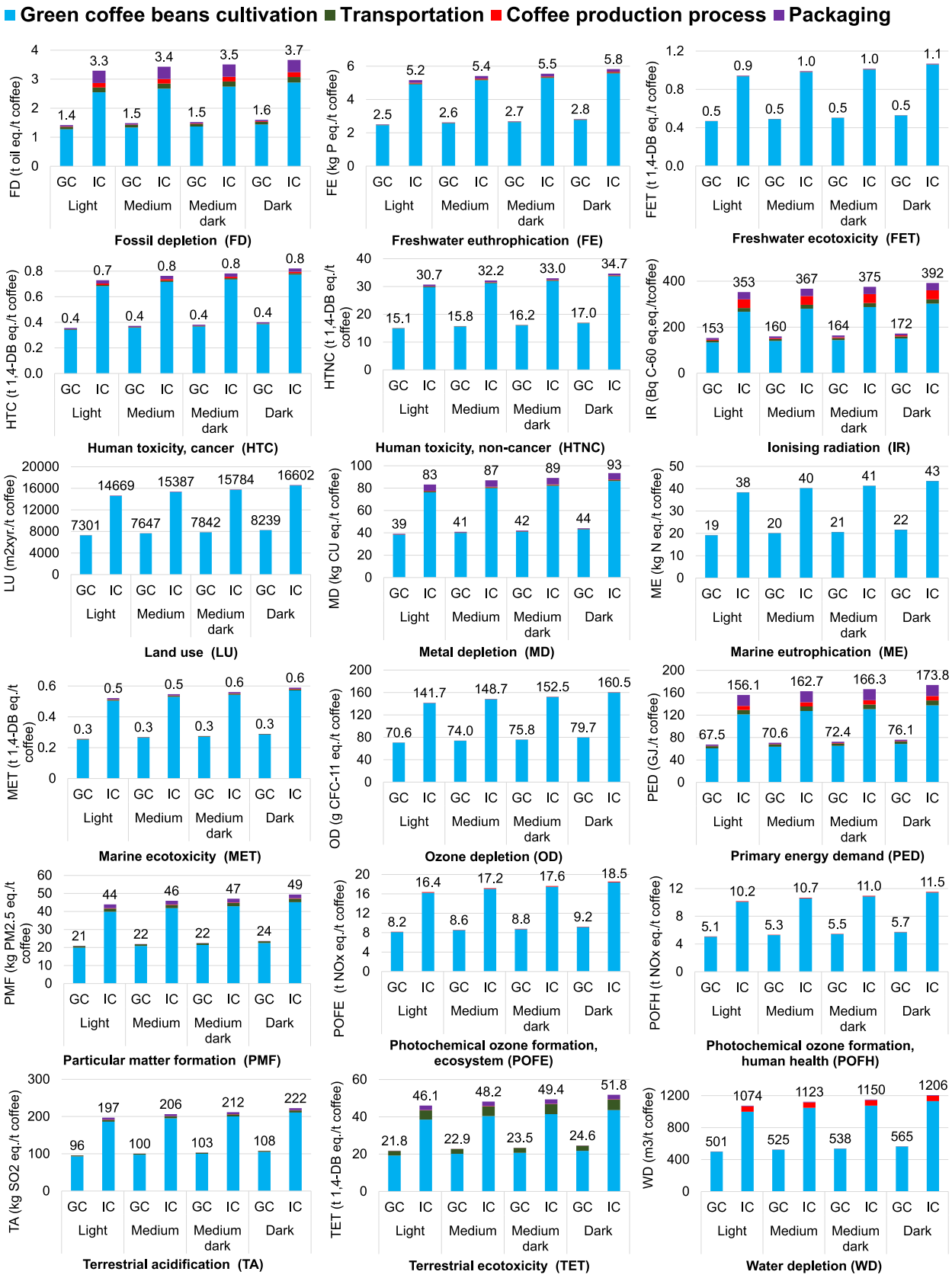


Fig. 11. Other cradle-to-gate impacts of ground and instant coffee production for different roasting levels (see Table S21 for further details). [GC = ground coffee; IC = freeze-dried instant coffee].

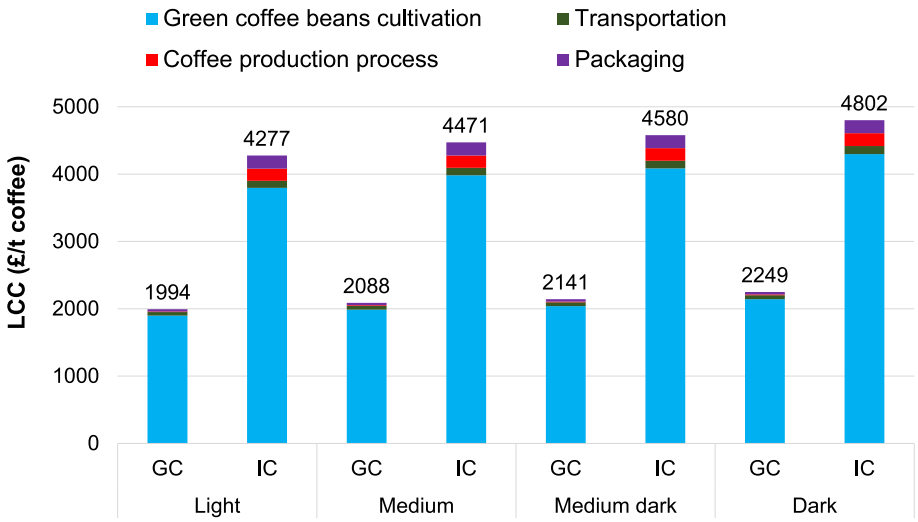


Fig. 12. Cradle-to-gate life cycle costs (LCC) of ground and instant coffee production for different roasting levels. [Scope: from cradle-to-gate].

costs of coffee production if the end-of-life treatment of SCGs are considered as they are a significant source of waste in instant coffee production. This has been carried out by expanding the system boundaries of both coffee products to include the end-of-life treatment of SCGs. For this sensitivity analysis, SCGs are assumed to be treated via incineration with energy recovery, which is a common practice in the coffee industry [59,60].

In the case of ground coffee, SCGs are generated during the consumption stage. Therefore, the amount of SCGs was estimated assuming a coffee extraction yield of 21.6%, considering the average brewing ratio for drip-filter coffee (50 g of ground coffee per one litre of water), based on the 'Brewing Control Chart' provided by the Specialty Coffee Association [76]. This gives a total of 3295.2 kg of SCGs per tonne of ground coffee with a 76.2% moisture content for all roasting levels. Based on the extraction-process mass balances, the amount of SCGs generated in the instant coffee production is equal to 4026 kg/t (Table 6), with a 76.1% moisture content. The energy recovery has been estimated based on the LHV of SCGs of 17.52 MJ/kg (dry basis) [77] as shown in Table S22. The system has been credited for the production of heat, assuming the displacement of an equivalent amount of natural gas with a LHV of 47.1 MJ/kg [78].

The results given in Tables S23 and S24 indicate that the incineration of SCGs leads to minor or negligible increase of CC (<2%) for both ground and instant coffee. In contrast, the release to water of heavy metals from the production and end-of-life treatment of incinerator equipment, increase FE, HTC, HTNC and MET by 4–9%, whilst the rest of the impacts are not affected when considering the treatment of SCGs. The LCC increase by 13–15% for ground and 7–8% for instant coffee, respectively, as the cost saving from the displacement of natural gas is not sufficient to compensate the additional costs of operating the incinerator (for the latter, see Table S22).

3.4. Sectoral and national analysis of coffee production

In addition to assessing the production of ground and instant coffee, this section includes the cradle-to-gate LCA and LCC results for coffee pods and instant-coffee mix to allow estimation of the impacts and costs at the sectoral and national levels. This is presented next, followed by the impacts and LCC of coffee production at the sectoral and national levels.

3.4.1. Coffee pods

The LCA and LCC results for coffee pods have been estimated by using the gate-to-gate results for ground coffee packaged in coffee pods presented in Section 3.2.3.1 and adding the impacts of green coffee bean cultivation. These results are presented in Figs. S5-S7 for CC, the other environmental impacts and LCC, respectively. As can be seen Fig. S5, the CC impact of coffee pods ranges from 7.6 to 8.5 t CO_2 eq./t, 8–9% higher than ground coffee in the laminated pouch. This is because of a higher amount of aluminium in the pods than in the pouches. Nevertheless, the packaging contributes only 8–9% to the total impact. The main contributor is cultivation of green coffee beans, causing 84% of CC. The contribution of the production process and transportation is relatively small (<4%).

The other environmental impacts of coffee pods are 3–109% higher than ground coffee for all impact categories, and in particularly for MD, HTC and IR which are 36–41%, 48–54%, and 97–109% higher, respectively (Fig. 11 and Fig. S6). The only exception to this is LU, POFE, POFH and OD where the environmental impacts of coffee pods are similar to ground coffee in pouches. Similar to the CC, the cultivation of green coffee beans is the main hotspot, contributing >62% to the impacts. The only exception is IR where the packaging is the main contributor (52–55%). The packaging also causes 12–37% of FD, FE, HTC, MD and PED. The contribution of transportation is relatively low for all of the impacts (<5%), except for TET (11%). The coffee production stage causes minor to negligible impacts (<4%).

The LCC of coffee pods are estimated at £2189-2443/t (Fig. S7). This is 9–10% higher than ground coffee packed in the pouch. Like the environmental impacts, the majority of the costs are from the cultivation of green coffee beans (87–88%), followed by packaging (9–10%). Transportation and the production process contribute <3% of the total cost.

3.4.2. Instant-coffee mix

Various types of instant coffee are available on the market, such as instant-coffee mix and ready-to-drink instant coffee. This section focuses on the former as the second most popular instant coffee product, after conventional instant coffee [4].

The environmental impacts of instant-coffee mix, estimated as mentioned in Section 2.2.2.5, are shown in Fig. S8 for CC and Fig. S9 for the other impacts. As indicated in Fig. S8, the CC impact is estimated at 6.9–7.1 t CO_2 eq./t, two times lower than for instant coffee. This is due to the much lower amount of coffee (11%) used in the product and replacement of glass packaging, despite the addition of significant amounts of sugar and milk powder. The latter is the main contributor (69–71%) to CC, followed by instant coffee (23–25%), packaging (3–4%) and sugar (3%).

The other environmental impacts of instant-coffee mix are also 2–9

times lower than for instant coffee for the same reasons. However, the contribution of coffee is much more significant for most (12) impacts, than for CC, contributing to 38–99% of the total impacts (Fig. S9), related to the cultivation of green coffee beans. Milk powder is the main hotspot for the remaining categories (PMF, IR, LU, ME, OD and TA), contributing 48–73% due to the production of animal feed.

The LCC of instant-coffee mix are in the range of £1945-2003/t (Fig. S10). This is 54–58% lower than the costs of instant coffee as milk powder and sugar are 5.3 and 1.7 times cheaper than coffee, respectively.

3.4.3. Environmental impacts and costs at the sectoral and national levels

A sectoral analysis considering instant, ground, coffee pods and instant-coffee mix has been carried out by extrapolating the environmental impacts and LCC estimated for ground and instant coffee presented previously. The results are presented in Figs. S11-S13 for different roasting levels assumed in the mix (from light roast dominating to dark roast being preferred).

As indicated in Fig. S11, the annual CC impact of coffee production ranges from 861 to 893 kt CO_2 eq./yr depending on the roasting levels assumed. This contributes 0.6% to the total GHG emissions from the UK's food and drink sector of 154.8 Mt CO_2 eq./yr [79][1]. The production of instant coffee has the largest contribution (70%) due to the highest demand and the highest CC impact among the other types of coffee. The next highest contributor is ground coffee with 16%, followed by coffee pods with 12% and instant-coffee mix with the remaining 2%.

The production of coffee consumes 8.7–9 PJ of the primary energy demand (Fig. S12), equivalent to 1% of the energy used in the UK food and drink sector [80]. The same contribution trend is observed as for CC. It is not possible to put the other impacts in context due to a lack of data at the sectoral level.

As shown in Fig. S13, the LCC of coffee production at the sectoral level ranges from £244 to 253 M/yr depending on the roasting level assumed in the market mix. Like CC, instant coffee has the largest contribution (69%) due to the highest production volume and the highest LCC of all types of coffee. Ground coffee adds another 16%, followed by ground coffee pods with 12% and instant-coffee mix with the remaining 3%. The contribution of the costs of coffee production to the total costs at the sectoral level cannot be determined due to lack of cost data for other types of coffee or other food products.

Comparing the different roasting level scenarios, the one assuming that the light roast dominates the market mix (SC3: 40% light, 30% medium, 20% medium-dark and 10% dark roast) has the lowest environmental impacts and LCC. However, the difference from the worst scenario (dark roast dominates) is small (3–4%). Nevertheless, switching to the light roast (i.e. from SC2 to SC3) could help reduce around 32 kt CO_2 eq./yr relative to the dark roasts. In addition, if silverskin is used as a fuel for roasting, the CC impact could be reduced by further 5 kt CO_2 eq./yr. A further reduction of 37 kt CO_2 eq. could also be achieved if ground coffee and instant coffee are packed in C/LDPE and laminated pouch packaging, respectively. These changes would be equivalent to a 5% reduction of GHG emissions from the coffee industry.

4. Conclusions

This work has integrated for the first time process design and simulation with life cycle assessment and life cycle costing to identify opportunities for improving the environmental and economic sustainability of coffee production. The focus has been on two main types of coffee available on the market – ground and instant – considering eight different products and four roast levels. Two system boundaries have been considered: gate-to-gate with the focus on the coffee manufacturing process, and cradle-to-gate to assess all stages involved in coffee production.

The results suggest that ground coffee is environmentally and economically more sustainable than instant coffee for both system boundaries, because it requires less green coffee beans and energy to produce the same amount of product. For the gate-to-gate boundary, CO_2 emissions from coffee roasting and packaging are the main hotspots for the CC impact for both coffee products. Packaging is also the dominant hotspot across most other environmental impact categories for both types of coffee. For the cradle-to-gate scope, the cultivation of green coffee beans is the main hotspot for all environmental impacts and costs as it contributes >84% to the impacts and costs of ground coffee and 75% of instant coffee. The results also indicate that light-roasted coffee has lower impacts and costs compared to the dark roast due to lower energy consumption and materials losses. For example, the CC impact and costs of light roast ground coffee are 12–13% lower than of the dark roast for both system boundaries. Therefore, the production of coffee with milder roasting levels can help improve the environmental and economic sustainability in the coffee industry.

The results also reveal that silverskin should be used for energy recovery rather than treating it as waste, as this can generate sufficient heat to be used within the process and reduce the gate-to-gate CC of ground and instant coffee by 7% and 3%, respectively. Furthermore, the C/LDPE and laminated pouch are the best packaging materials for ground and instant coffee, respectively, as they have the lowest environmental impacts and costs compared to the other materials considered.

Based on the annual production of coffee at the sectoral level, the CC impact is estimated to be 861–893 kt CO_2 eq./yr depending on the roasting level. This is equivalent to up to 1.3% of the total GHG emissions from UK's food sector. If light roasts are used, energy is recovered from silverskin and the packaging is changed to C/LDPE and laminated pouches, the emissions and costs could be reduced by 42 kt CO_2 eq./yr and £10 M/yr. This reduction is equivalent to a 5% reduction of emissions from the coffee industry.

Future work should expand the system boundaries of coffee to include the consumption stage and to explore different valorisation and/or treatment options for SCG. This is in part to avoid burden shifting across other life cycle stages, e.g. the energy consumption associated with different coffee preparation methods. Future efforts should also look at developing data on mass transfer properties, e.g. solubility, for coffee to support robust modelling of such systems for further improvement in the inventory data for the extraction process.

It is expected that the findings from this work will be of interest to coffee producers and consumers, as well as policy makers, seeking to improve the environmental and economic sustainability of the coffee industry.

5. Financial sources

This work was supported by the Royal Thai Government and EPSRC [Grant number EP/K011820/1]. The authors acknowledge gratefully this funding.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

[1] Includes emissions associated with land-use change of imported products that are most likely to be linked to drivers of tropical deforestation, such as coffee, cocoa, beef, pork, poultry, oilseeds, etc.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2023.142793.

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